ZapBox Duo – Operating Instructions

Language: English | **Version:** OI939600

Overview

The **ZapBox Duo** is an electronic switch for Bitcoin Lightning payments. A payment over the Lightning Network triggers two independent outputs – ideal for vending machines, presentations, event control, and many other applications.

Standard Equipment

Component	Description
Microcontroller	T-Display-S3 with 1.9" LCD display
Front panel	Optional 35° or 90° angle
Input	USB-C (Power IN)
Output Channel 1 (CH1)	30 A power relay with external terminals
Output Channel 2 (CH2)	Dual socket with USB-A and USB-C
Control element	LED button (full ZapBox operability)
Switch 1	3-position slide switch (AUTO / OFF / ON)
Switch 2	2-position slide switch (Invert Output)
BTC ticker	Optional
Board buttons	2 on-board micro buttons (optional)
NFC module	For Boltcards (optional)

Views

Image 1: Front view

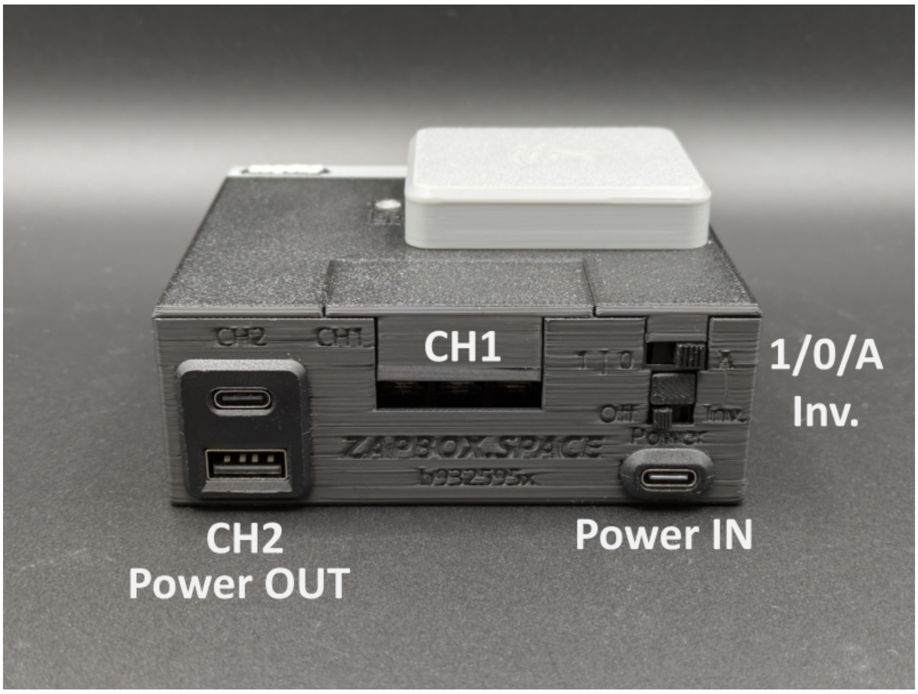

Image 2: Rear view

Getting Started

Power Supply

Connect the device to a power source via USB-C cable at the **Power IN** port with **5 V DC (max. 5 A)**.

> **Note:** The Power IN port is not "intelligent". Some chargers or power modules with a USB-C output do not recognise the ZapBox and will not supply power. In this case, use a **USB-A output** of your power supply or a different power source.

First Test (pre-configured device)

If the device is already configured, you can run a quick first test:

1. Press the **LED button** to cycle through the display screens.
2. Scan the displayed **QR code** with a Lightning wallet.
3. Pay the invoice.
4. You should hear the relay click – the switching operation was successful.

Setup

Web Installer

For initial setup and firmware updates, use the **Web Installer**:

https://installer.zapbox.space/

The full setup process is described there. It is recommended to always flash the **"Latest" firmware** version to keep the ZapBox up to date.

USB Data Access (hidden panel)

To read data from or transfer data to the device, connect the ZapBox to a computer or laptop:

1. On the **right side of the front panel** there is a small hidden flap.
2. Open the flap by pushing it to the right with a **thin flat-head screwdriver** inserted from below.
3. Connect a USB-C cable to the microcontroller port behind the flap.

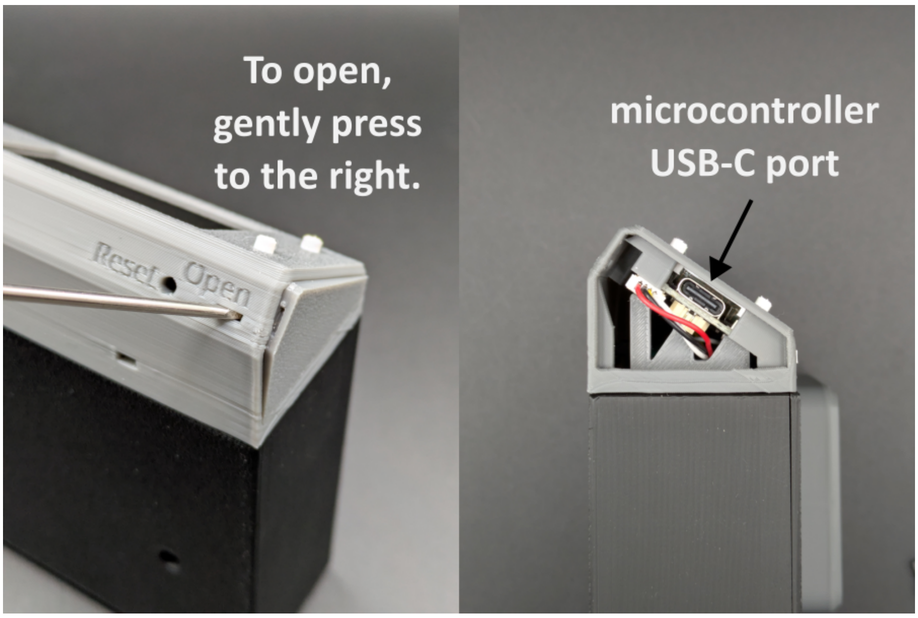

Image: Opening the panel and USB-C data port

> **Important:** The USB port on the microcontroller is exclusively intended for flashing new firmware or transferring configuration parameters. If switching functions are triggered simultaneously during flashing, this can cause **malfunctions or damage to the microcontroller**.
>
> It is therefore strongly recommended to either:

> - Avoid triggering any switching functions while flashing, **or**
> - additionally connect the regular **Power IN** port to the same power supply, so that current for the power relay does not flow through the microcontroller and overload it.

Slide Switches

The ZapBox Duo has two slide switches.

Switch 1 – 3-Position Slide Switch (AUTO / OFF / ON)

Position	Function
A (AUTO)	Automatic mode – normal operation
0 (OFF)	Power supply disconnected – output OFF
1 (ON)	Output CH2 (dual USB A/C) permanently ON

Switch 2 – 2-Position Slide Switch (Invert / Normal)

Position	Function
Normal	CH2 is de-energised (0 V) at rest. After switching, 5 V is present at the USB sockets.
Inv. (Invert)	CH2 is at 5 V at rest. After switching, the output changes to 0 V (inverse switching).

> **Note:** If the 2-position switch is set to **Inv.** and the 3-position switch is set to **position 1**, the output is **OFF** – contrary to normal operation where it would be ON.

Outputs

Channel 1 (CH1) – 30 A Power Relay

The power relay on Channel 1 exposes the contacts **COM / NO / NC** on the outside. These are factory-covered with a **protective cap**.

- **Removing the cap:** Simply pull it upward.
- **Replacing the cap:** Place the long side first, then press the short angled side downward until the cap snaps into place.

The power relay contacts are rated for a **maximum current of 30 A**.

On the **top of the ZapBox** there is a small transparent opening. An LED behind it indicates the **ON status of the power relay**.

Channel 2 (CH2) – Dual USB Socket (USB-A / USB-C)

The dual USB socket is switched via a relay switching contact (CH2). The **total load** on the sockets should **not exceed 3 A**.

NFC Module (optional)

Depending on the configuration, an NFC module is mounted on the **top of the ZapBox**. It supports the following card types:

- **Boltcards** (NTAG424)
- **LNURL-Withdraw** from NTAG21x (213 / 215 / 216)

Controls

Depending on the version, the ZapBox is equipped with two small **on-board micro buttons** in addition to the LED button, connected directly to the microcontroller. All functions can be accessed via both the LED button and the micro buttons.

Function Overview

Function	Micro button	LED button
Show Help page	Press HELP 1×	Hold LED button for at least 2 sec.
Next page / product change	Press NEXT 1×	Press LED button 1× briefly
Show REPORT page	Press HELP 2×	Press LED button 3× in quick succession
Enter Config mode	Hold NEXT for at least 5 sec.	Press 1× briefly, then hold for at least 5 sec.

Technical Specifications

Property	Value
Supply voltage	5 V DC via USB-C
Maximum input current	5 A
CH1 switching capacity	max. 30 A
CH2 output capacity	max. 3 A (total)
Display	1.9" LCD (T-Display-S3)
Communication	Wi-Fi (ESP32-S3)
Payment protocol	Bitcoin Lightning Network

Safety Instructions

- Only operate the device with the specified supply voltage.
- Do not exceed the maximum current ratings of the outputs.
- Do not perform any work on the relay contacts under load.
- The device is not suitable for use in humid or wet environments.
- Keep out of reach of children.

Further Resources

Resource	Link
Overview of all ZapBox models	https://zapbox.space/
Web Installer, quick reference & troubleshooting	https://installer.zapbox.space/
Detailed documentation (parameters & functions)	https://ereignishorizont.xyz/zapbox/
GitHub repository (software, PCB layouts, 3D print files)	https://github.com/AxelHamburch/ZapBox

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Subject to change without notice. As of 2026